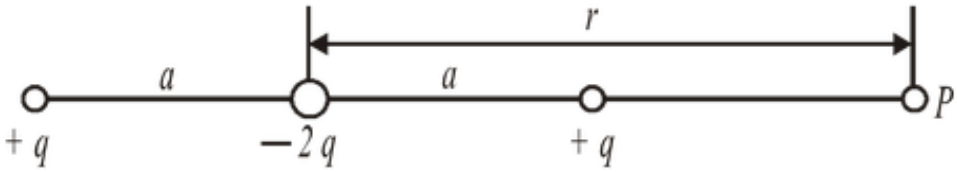
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Determine the electric field intensity at point P due to quadruple distribution shown in figure for $r \gg a$



(A) 0

(B) kqa^2/r^4

(C) $6\mathrm{kq}a^2/\mathrm{r}^4$

(D) $6kqa^2/r^2$

Q.2

A positively charged ball hangs from a long silk thread. Electric field at a certain point (at the same horizontal level of the ball) due to this charge is E. Let us put a positive test charge q_0 at this point and measure F/q_0 on this charge. Then, E

(A) $> F/q_0$

(B) $< F/q_0$

(C) $= F/q_0$

(D) none of these

Q.3

An electric dipole of moment $\vec{p}$ is placed at the origin along the X-axis. The electric field at a point P, whose position vector makes an angle θ with the X-axis, will make an angle φ with the X-axis, then φ is ($\alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right)$)

(A) $\phi = \theta + 3\alpha$

(B) $\phi = \theta + 2\alpha$

(C) $\phi = \theta + \alpha$

(D) $\phi = \theta$

Q.4

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Three +ve charges of equal magnitude ' q ' are placed at the vertices of an equilateral triangle of side I. How can the system of charges be placed in equilibrium ?

(A) $Q = \left(-\frac{q}{\sqrt{3}}\right)$ at the centroid of the triangle

(B) By placing a charge $Q = \left(\frac{q}{\sqrt{3}}\right)$ at the centroid of the triangle

(C) By placing a charge $Q = q$ at a distance $/$ from all the three charges

(D) By placing a charge $Q = -q$ above the plane of the triangle at a distance $/$ from all the three charges

Q.5

A conducting loop of negligible thickness and of radius $a = 6\ \text{cm}$, carries a charge of $q = 4\mu\text{C}$. The electric field (E) on the axis of symmetry of the loop at a distance of $x = 8\ \text{cm}$ from the centre of loop is

(A) $288 \times 10^3 \text{NC}^{-1}$

(B) $288 \times 10^4 \text{NC}^{-1}$

(C) $288 \times 10^5 \text{NC}^{-1}$

(D) $288 \times 10^2 \text{NC}^{-1}$

Q.6

Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. The net charge on the sphere is then

(A) Negative and distributed uniformly over the surface of the sphere

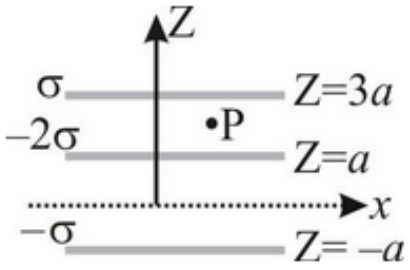
(B) Negative and appears only at the point on the sphere closest to the point charge

(C) Negative and distributed non-uniformly over the entire surface of the sphere

(D) Zero

Q.7

Three infinitely long charge sheets are placed as shown in figure. The electric field at point P is



(A) $\frac{2\sigma}{\varepsilon_0}\hat{k}$

(B) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$

(C) $\frac{4\sigma}{\varepsilon_0}\hat{k}$

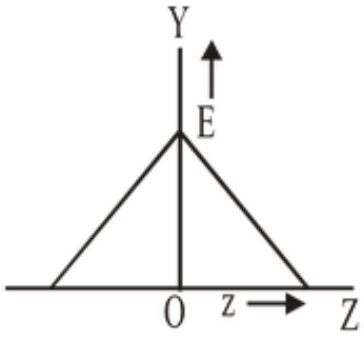
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(D) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$

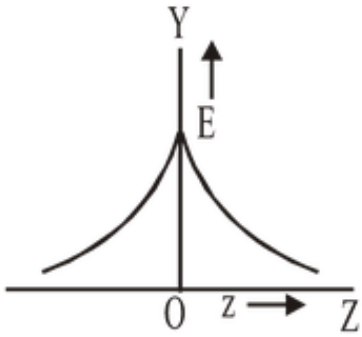
Q.8

A circular ring carries a uniformly distributed positive charge and lies in $X-Y$ plane with center at origin of co-ordinate system. If at a point $(0,0,z)$ the electric field is E, then which of the following graphs is correct?

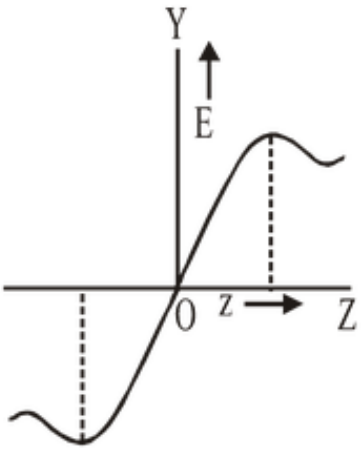
(A)



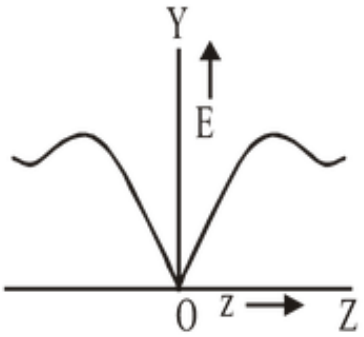
(B)



(C)



(D)

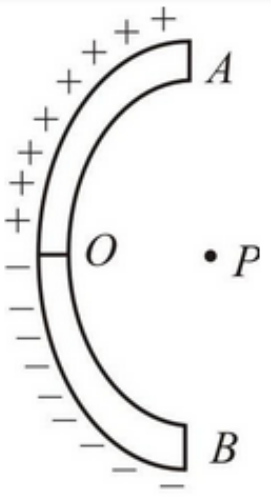


Q.9

A thin glass rod is bent into a semicircle of radius r. A charge $+Q$ is uniformly distributed along the upper half and a charge $-Q$ is uniformly distributed along the lower half, as shown in figure. Calculate electric field

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E at p, the center of semicircle.



(A) $\frac{Q}{\pi^2\varepsilon_0 r^2}$

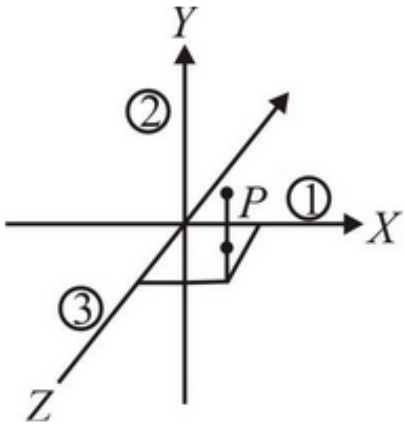
(B) $\frac{2Q}{\pi^2\varepsilon_0 r^2}$

(C) $\frac{4Q}{\pi^2\varepsilon_0 r^2}$

(D) $\frac{Q}{4\pi^2\varepsilon_0 r^2}$

Q.10

Find the electric field vector at $P(a, a, a)$ due to three infinitely long lines of charges along $x-$, $y-$ and z - axes respectively. The charge density, i.e., charge per unit length of each wire is λ.



(A) $\frac{\lambda}{3\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$

(B) $\frac{\lambda}{2\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$

(C) $\frac{\lambda}{2\sqrt{2}\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$

(D) $\frac{\sqrt{2}\lambda}{\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$

Q.11

Three identical spheres, each having a change q and radius R, are kept in such a way that each touches the other two. Find the magnitude of the electric force on any sphere due to other two

(A) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{R}\right)^2$

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(B) $\frac{\sqrt{3}}{4\pi\varepsilon_0}\left(\frac{q}{R}\right)^2$

(C) $\frac{\sqrt{3}}{16\pi\varepsilon_0}\left(\frac{q}{R}\right)^2$

(D) $\frac{\sqrt{5}}{16\pi\varepsilon_0}\left(\frac{q}{R}\right)^2$

Q.12

Three concentric spherical metallic shells A, B and C of radii a, b and $c(a < b < c)$ have charge densities $\sigma, -\sigma$ and σ respectively. The potential of A, B and C are

(A) $V_A = \frac{1}{\varepsilon_0}(a+b+c)\sigma$

(B) $V_B = \frac{1}{\varepsilon_0}\left(\frac{a^2}{b} - b + c\right)\sigma$

(C) $V_C = \frac{1}{\varepsilon_0}\left(\frac{a^2}{c} + \frac{b^2}{c} + c\right)\sigma$

(D) $V_A = V_B = V_C = \frac{1}{\varepsilon_0}(a+b+c)\sigma$

Q.13

Consider a dipole of moment p placed in an uniform electrostatic field E_0. The radius of the corresponding equipotential surface is

(A) $(kp/\mathrm{E}_0)^{1/3}$

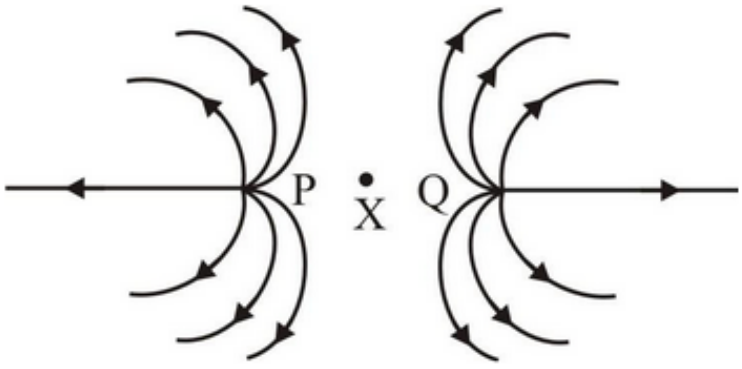
(B) kp/E_0

(C) $(kp/\mathrm{E}_0)^{2/3}$

(D) none of these

Q.14

The diagram below shows electric field lines around two isolated point changes P and Q. At X the field strength is zero. Which of the following is true ?



(A) Q is smaller charge than P because X is closer to P than Q

(B) Field strength is always proportional to the distance from X

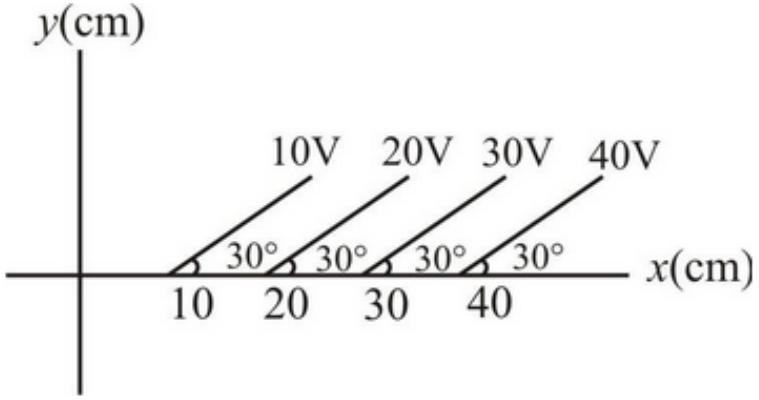
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(C) The potential at Q is less than the potential at P

(D) The field lines show that charges are positive

Q.15

Electric field represented by equipotential surface shown in figure is



(A) $\vec{E} = 200(\hat{i} + \sqrt{3}\hat{j})NC^{-1}$

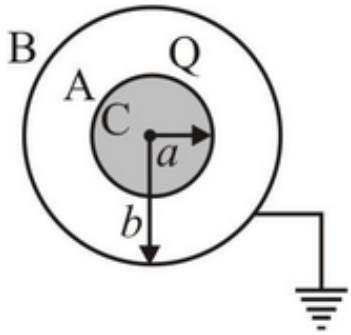
(B) $\vec{E} = 100(\hat{i} + \sqrt{2}\hat{j})NC^{-}{}_{1}$

(C) $\vec{E} = 100(-\hat{i} + \sqrt{3}\hat{j})NC^{-1}$

(D) $\vec{E} = 200(-\hat{i} + \sqrt{3}\hat{j})NC^{-1}$

Q.16

A conducting sphere of radius a, with charge Q, is placed concentrically inside a positive conducting shell B of or radius b. B is earthed. C is the common centre of A and B. Study the following statements



i. The potential at a distance r from C, where $a \leq r \leq b$, is $\frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$

ii. The potential difference between A and B is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{a} - \frac{1}{b}\right)$

iii. The potential at a distance r from C, where $a \leq r \leq b$, is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{r} - \frac{1}{b}\right)$

Which of the following statements are correct?

(A) Only (i) and (ii)

(B) Only (ii) and (iii)

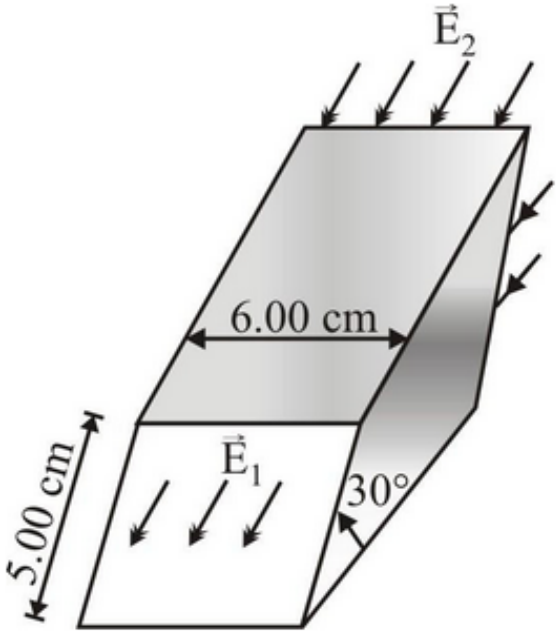
(C) Only (i) and (iii)

(D) All

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Q.17

The electric field $\vec{E}_1$ at one face of a parallelopiped is uniform over the entire face and is directed out of the face. At opposite face, the electric field $\vec{E}_2$ is also uniform over the entire face and is directed into that face (as shown in figure). The two faces in question are inclined at 30° from the horizontal, while $\overrightarrow{E_1}$ and $\overrightarrow{E_2}$ (both horizontal) have magnitudes of 2.50×10^4 N/C and 7.00×10^4 N.C. Assuming that no other electric field lines cross the surfaces of the parallelopiped, determine the net charge contained within.



(A) $-67.5\varepsilon_0 \mathrm{C}$

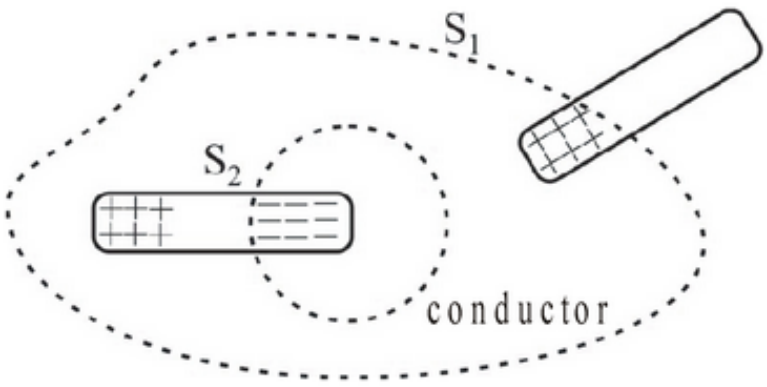
(B) $37.5\varepsilon_0 \mathrm{C}$

(C) $105\varepsilon_0 \mathrm{C}$

(D) $-105\varepsilon_0 \mathrm{C}$

Q.18

Charge on an originally uncharged conductor is separated by holding a positively charged rod very nearby, as in figure Assume that the induced negative charge on the conductor is equal to the positive charge q on the rod. Then, flux through surface S_1 is



(A) zero

(B) q/ε_0

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(C) $-q/\varepsilon_0$

(D) none of these

Q.19

A flat, square surface with sides of length L is described by the equations
$x = L, 0 \leq y \leq L, 0 \leq z \leq L$
Find the electric flux through the square due to a positive point charge q located at the origin $(x = 0, y = 0, z = 0)$

(A) $\frac{q}{4\varepsilon_0}$

(B) $\frac{q}{6\varepsilon_0}$

(C) $\frac{q}{24\varepsilon_0}$

(D) $\frac{q}{48\varepsilon_0}$

Q.20

An electric dipole is put in a north-south direction in a sphere filled with water. Which statements is correct ?

(A) Electric flux is coming towards sphere

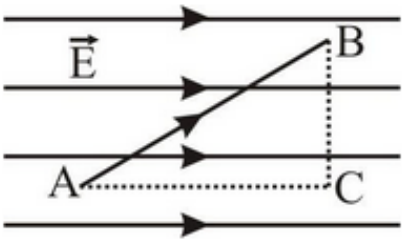
(B) Electric flux is coming out of sphere

(C) Electric flux entering into sphere and leaving the sphere are same

(D) Water does not permit electric flux to enter into sphere

Q.21

An electron is taken from a point A to point B along the path AB in a uniform electric field of intensity $E = 10\text{Vm}^{-1}$, Side $AB = 5$ m and side $BC = 3$ m. Then, the amount of work done is (Mark answer in eV)



Q.22

An infinite number of charge, each of charge $1\mu\text{C}$, are placed on the x-axis with co-ordinates $x = 1, 2, 4, 8, \ldots \infty$. If a charge of $1C$ is kept at the origin, then what is the net force acting on 1C charge in kN (kilonewton)

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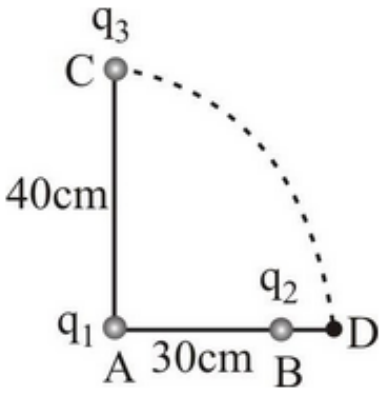
Q.23

A non-conducting ring of radius $0.5\ \mathrm{m}$ carries a total charge of $1.11 \times 10^{-10}\mathrm{C}$ distributed non-uniformly on its circumference producing an electric field $\vec{E}$ everywhere in space.

The value of the line integral $\int_{l=\infty}^{l=0} -\vec{E} \cdot \overrightarrow{dl}(l, = 0$ being centre of the ring) in volt is

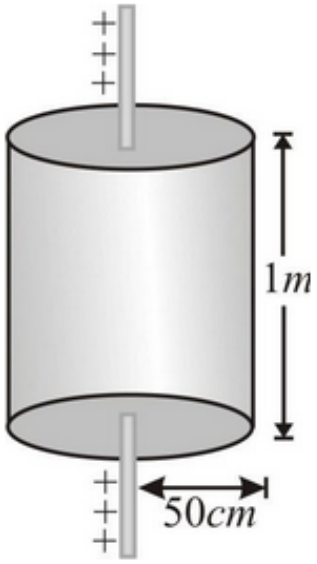
Q.24

Two charges q_1 and q_2 are placed 30 cm apart, as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius $40\ \mathrm{cm}$ from C to D. The change in the potential energy of the system is $\frac{q_3}{4\pi\varepsilon_0}k$, where k is (take $\mathrm{q_2} = 1\mathrm{C}$)



Q.25

Electric charge is uniformly distributed along a long straight wire of radius $1\ \mathrm{mm}$. The charge per cm length of the wire is Q coulomb. Another cylindrical surface of radius 50 cm and length $1\ \mathrm{m}$ symmetrically encloses the wire as shown in the figure. The total electric flux passing through the cylindrical surface is $\mathrm{K}\frac{Q}{\varepsilon_0}$, where K is a positive integer. Find K.



Q.26

An electric field is expressed as $\vec{E} = 2\hat{i} + 3\hat{j}$. Find the potential difference $(V_A - V_B)$ between two points

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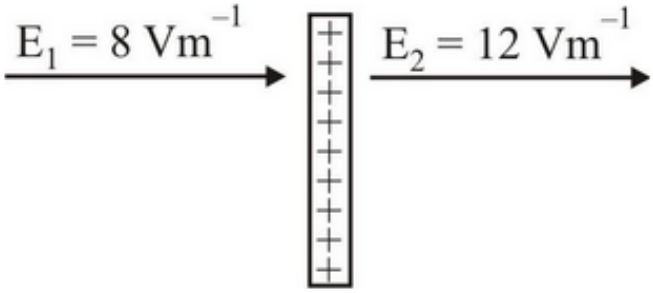
A and B whose position vectors are given by $r_A = \hat{i} + 2\hat{j}$ and $r_B = 2\hat{i} + \hat{j} + 3\hat{k}$. Mark modulus of potential difference.

Q.27

Two conducting spheres, one of radius $6\ \text{cm}$ and the other of radius $12\ \text{cm}$, each carrying $3 \times 10^{-8}\text{C}$, are placed very far apart. If the spheres are connected by a conducting wire, the magnitude of the charge transferred is K. Find $\text{K} \times 10^8$.

Q.28

The electric field on two sides of a large charged plate is shown in Fig. The charge density on the plate in S.I. Units are given by $K\varepsilon_0$ (ε_0 is the permittivity of free space in S.I. Units), Find value of K

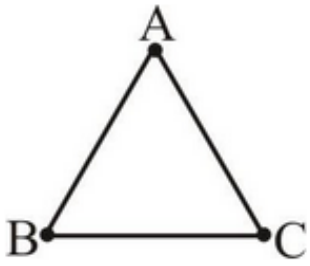


Q.29

The electric intensity due to an infinite cylinder of radius R and having charge q per unit length at a distance $r(r > R)$ from its axis is inversely proportional to r^n, where n is a positive integer. Find n.

Q.30

Three identical charge each of $2\mu\text{C}$ are placed at the vertices of a triangle ABC as shown in the figure



If $AB + AC = 12\ \text{cm}$ and $AB \cdot AC = 32\ \text{cm}^2$, the potential energy of the charge at A in Joules is

Answers & Solutions

Q.1 Answer:

$6\text{kq}a^2/\text{r}^4$

Solution:

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The field at point P is superposition of field $\vec{E}_1, \vec{E}_2, \vec{E}_3$ due to each charge.

$$\overrightarrow{E}_1 = \frac{q}{4\pi\varepsilon_0(r-a)^2}\hat{i}$$

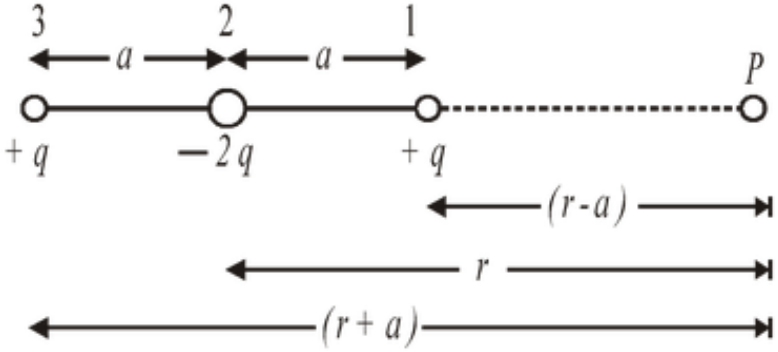
$$\overrightarrow{E}_2 = \frac{2q}{4\pi\varepsilon_0 r^2}\hat{i}, \quad \overrightarrow{E}_3 = \frac{q}{4\pi\varepsilon_0(r+a)^2}\hat{i}$$

Now,

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-a)^2} - \frac{2}{r^2} + \frac{1}{(r+a)^2}\right]^2 \hat{i}$$

$$E = \frac{q}{4\pi\varepsilon_0 r^2}\left[\left\{1 - \left(\frac{a}{r}\right)^{-2}\right\} - 2 + \left\{1 + \left(\frac{a}{r}\right)\right\}^{-2}\right]$$



If $r >> a$, we can use binomial approximation :

$$(1+\alpha)^n = 1 + n\alpha + \frac{n(n-1)}{2!}\alpha^2 + \ldots\ldots \text{ for } \alpha << 1\Big)$$

Therefore,

$$E = \frac{q}{4\pi\varepsilon_0 r^2}\left[\begin{array}{l}\left\{\left(1 - 2\left(-\frac{a}{r}\right)\right) + \frac{-2(-2-1)}{2}\left(\frac{-a}{r}\right)^2\right\} \\ -2 + \left\{1 - 2\frac{a}{r} + \frac{-2(-2-1)}{2}\left(\frac{a}{r}\right)^2\right\}\end{array}\right]$$

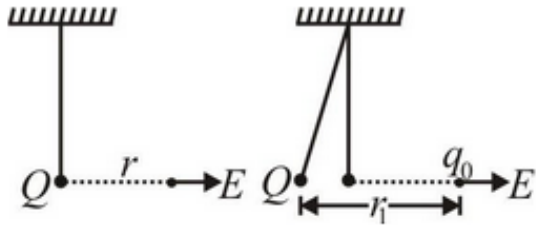
$$= \frac{6a^2 q}{4\pi\varepsilon_0 r^4} = \frac{6kqa^2}{r^4}$$

Q.2 Answer:

$> F/q_0$

Solution:

$E = \frac{kQ}{r^2}$ **(i)**



$F = \frac{kQq_0}{r_1^2} \Rightarrow \frac{F}{q_0} = \frac{kQ}{r_1^2}$ **(ii)**

As $r_1 > r$, so from (i) and (ii) $E > \frac{F}{q_0}$

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Q.3 Answer:

$\phi = \theta + \alpha$

Solution:

An electric dipole of moment $= \bar{p}$

electric field x-axis at a point $= \mathrm{p}$

angle $= \theta$ **with** $x-$ **axis**

$\tan a = \frac{1}{2}\tan\theta$

$(\theta + \alpha) =$ **the value of the position vector makes an angle** θ

$\theta = 60^\circ + \alpha$

now resolving E **into its components**

$E\cos\alpha = \frac{2p\cos 60^\circ}{4\pi\epsilon_0 r^3} \rightarrow (1)$

$E\sin\alpha = \frac{p\sin 60^\circ}{4\pi\epsilon_0 r^3} \quad \rightarrow (2)$

Dividing 2 by 1

$\tan\alpha = \frac{1}{2}\tan\theta$

$\tan\alpha = \tan(\theta + \alpha)$

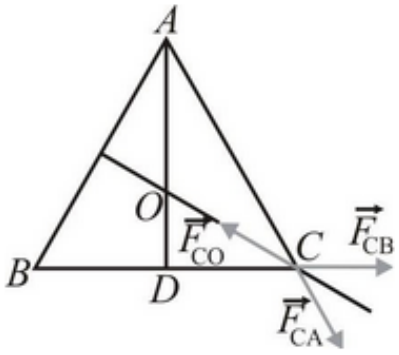
$\alpha = \theta + \alpha$

Q.4 Answer:

$Q = \left(-\frac{q}{\sqrt{3}}\right)$ at the centroid of the triangle

Solution:

To keep the system in equilibrium, net force experienced by charges at ' A', $'B'$ **and '** C' **should be zero. For this another charge of the opposite sign should be placed at the centroid of the triangle. Let this charge be '** $-Q'$.



$AD = l\cos 30^\circ = \frac{l\sqrt{3}}{2}, AO = \frac{2}{3}AD = \frac{1}{\sqrt{3}}$

$2\left|\overrightarrow{F_{CA}}\right|\cos 30^\circ = \left|\overrightarrow{F_{CO}}\right|$

$2 \times \frac{1}{4\pi\varepsilon_0} \times \frac{q^2}{l^2} \times \frac{\sqrt{3}}{2} = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{(l/\sqrt{3})^2}$

$\Rightarrow Q = \frac{q}{\sqrt{3}}$

Q.5 Answer:

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$288 \times 10^4 \mathrm{NC}^{-1}$

Solution:

$E = \frac{kqx}{(a^2+x^2)^{3/2}} = 288 \times 10^4 N - C^{-1}$

Here $k = 9 \times 10^9\ \mathrm{N} - \mathrm{m}^2 - \mathrm{coul}^{-2}$.

Q.6 Answer:

Zero

Solution:

There is no change in net charge due to induced charges. Net charge is conserved and it will be zero. But at near point charge sphere have induced non-uniform negative charge and the away part of the sphere has induced non-uniform positive charge. But the sum of these non-uniform negative and non-uniform positive is equal to zero.

Q.7 Answer:

$-\frac{2\sigma}{\varepsilon_0}\hat{k}$

Solution:

Electric field due to the charged sheet is $\mathrm{E} = \frac{\sigma}{2\epsilon 0}$

Away from the if the charge is positive and towards the sheet if the charge is negative.

Let the sheet are A, B, C

Electric field at a point in between the sheets B and C is :

Electric field due to A **is** $\mathrm{E} = \frac{\sigma}{2\epsilon 0}$ **towards** A

electric field due to B **towards** B **is** $2\left(\mathrm{E} = \frac{\sigma}{2\epsilon 0}\right)$

Therefore, electric field due to C **is** $\mathrm{E} = \frac{\sigma}{2\epsilon 0}$ **away from** C

Net electric field

$= \frac{\sigma}{2\varepsilon_0}(1+2+1) = 4\left(\frac{\sigma}{2\varepsilon_0}\right)$

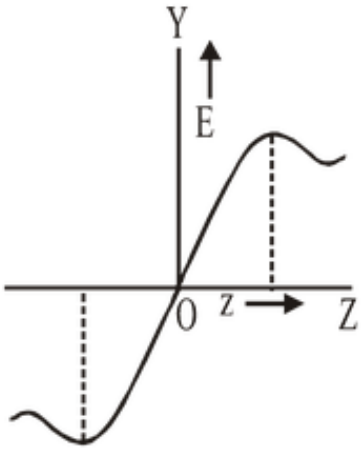
$= \frac{2\sigma}{\varepsilon_0}$

And along the z-axis along unit vector $\hat{k}$

$\frac{-2\sigma}{\varepsilon_0}\hat{k}$

Q.8 Answer:

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Solution:

Following two arguments shall lead us to the right choice.

(i) Electric field at the center of the ring is zero.

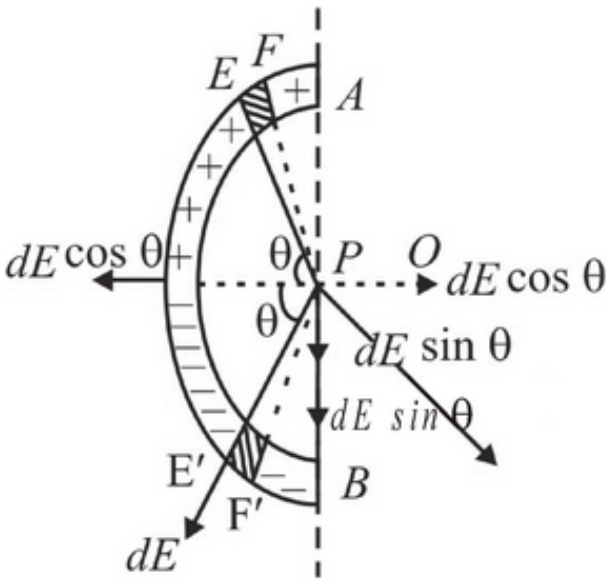
(ii) Electric field is directed away from the ring.

Q.9 Answer:

$\frac{Q}{\pi^2\varepsilon_0 r^2}$

Solution:

Take PO as the x-axis and PA as the y-axis. Consider two elements EF and $E'F'$ of width $d\theta$ at angular distance q above and below PO, respectively. The magnitude of the field at P due to either element is



$$dE = \frac{1}{4\pi\varepsilon_0}\frac{rd\theta \times \frac{Q}{\pi r/2}}{r^2} = \frac{Q}{2\pi^2\varepsilon_0 r^2}d\theta$$

Resolving the fields, we find that the components along PO sum up to zero and hence, the resultant field is along PB.

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$\therefore$ **Field at P due to pair of elements** $= 2dE\sin\theta$

$$E = \int_0^{\pi/2} 2dE\sin\theta$$

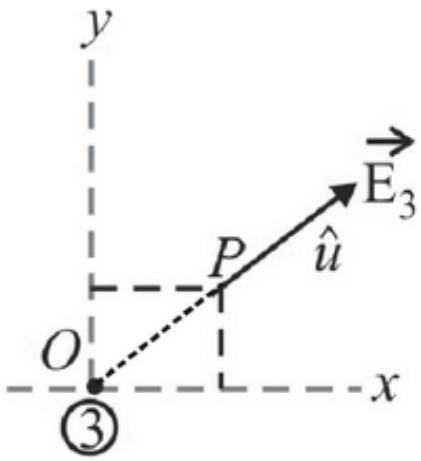
$$= 2\int_0^{\pi/2} \frac{Q}{2\pi^2\varepsilon_0 r^2}\sin\theta d\theta = \frac{Q}{\pi^2\varepsilon_0 r^2}$$

Q.10 Answer:

$\frac{\lambda}{2\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$

Solution:

Let us consider the electric field due to wire (3) only.



$$\overrightarrow{E_3} = E\hat{u}$$

$$\vec{E}_3 = \frac{\lambda}{2\pi\varepsilon_0(a^2+a^2)^{1/2}}\left(\hat{i}\cos 45^\circ + \hat{j}\cos 45^\circ\right)$$

$$= \frac{\lambda}{2\sqrt{2}\pi\varepsilon_o a}\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$$

$$\vec{E}_3 = \frac{\lambda}{4\pi\varepsilon_o a}(\hat{i}+\hat{j})$$

Similarly, electric field due to wires (1) and (2)

$$\vec{E}_1 = \frac{\lambda}{4\pi\varepsilon_0 a}(\hat{j}+\hat{k})_{\text{and}} \quad \overrightarrow{E_2} = \frac{\lambda}{4\pi\varepsilon_0 a}(\hat{i}+\hat{k})$$

$$\vec{E}_{\text{net}} = \overrightarrow{E_1} + \overrightarrow{E_2} + \overrightarrow{E_3}$$

$$\vec{E}_{\text{net}} = \frac{\lambda}{2\pi\varepsilon_0 a}(\hat{i}+\hat{j}+\hat{k})$$

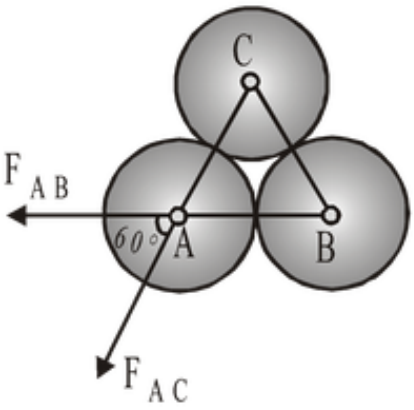
Q.11 Answer:

$\frac{\sqrt{3}}{16\pi\varepsilon_0}\left(\frac{q}{R}\right)^2$

Solution:

For external points, a charged sphere behaves as if the whole of its charge were concentrated at its center.

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Force on A due to B,

$F_{AB} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2R)^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{4R^2}$ **along BA**

And force on A due to C

$F_{AC} = \frac{1}{4\pi\varepsilon_0(2R)^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{4R^2}$ along CA

Now as angle between BA and CA is 60° and

$|F_{AB}| = |F_{AC}| = F$

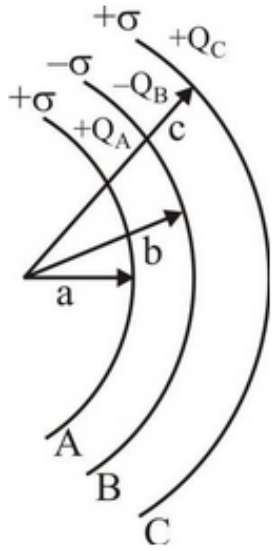
$\therefore \quad F_A = \sqrt{F^2 + F^2 + 2F \cdot F \cdot \cos 60} = \sqrt{3}F$

$F_A = \frac{1}{4\pi\varepsilon_0}\frac{\sqrt{3}}{4}\left(\frac{q}{R}\right)^2$

Q.12 Answer:

$V_B = \frac{1}{\varepsilon_0}\left(\frac{a^2}{b} - b + c\right)\sigma$

Solution:



$\frac{Q_A}{4\pi a^2} = \sigma$

$-\frac{Q_B}{4\pi b^2} = -\sigma$

$-\frac{Q_C}{4\pi c^2} = \sigma$

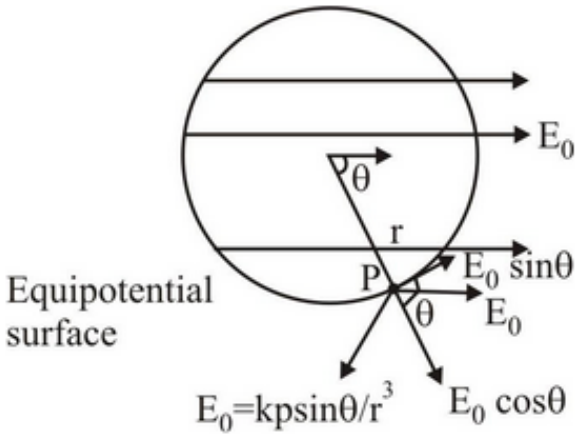
$V_B = \frac{Q_A}{4\pi\varepsilon_0 b} - \frac{Q_B}{4\pi\varepsilon_0 b} + \frac{Q_C}{4\pi\varepsilon_0 c} = \frac{1}{\varepsilon_0}\left(\frac{a^2}{b} - b + c\right)$

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Q.13 Answer:

$(kp/\mathrm{E}_0)^{1/3}$

Solution:



The radius of the equipotential surface is given by $\frac{kp\sin\theta}{r^3} = E_0\sin\theta \ldots (1)$

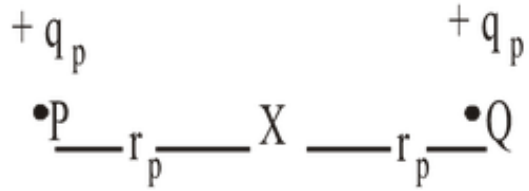
$r = \left(\frac{kp}{E_0}\right)^{1/3}$

where $\mathrm{k} = (4\pi\varepsilon_0)^{-1}$**. Equation (1) follows from the fact that on an equipotential surface the field is at right angles. So, the total tangential component is zero.**

Q.14 Answer:

The field lines show that charges are positive

Solution:



From the diagram, we conclude that (1) Q **is larger charge because** X **is closer to** P **than Q. When**

$E_P = E_Q = \frac{q_P}{4\pi\varepsilon_0 r_P^2} = \frac{q_Q}{4\pi\varepsilon_0 r_Q^2} \therefore \frac{q_P}{q_Q} = \left(\frac{r_P}{r_Q}\right)^2$

Since $r_Q > r_p$**, then** $q_Q > q_p .(2)$ **The field strength is always proportional to the inverse squared distance from** X **as it can be readily seen from the expression in** $(1), (3)$**. The potential at** P **or** Q **is infinitely large since the potential at the location of an isolated charge is theoretically infinite. (4) Both charges are positive since the field lines are radially outward from the charges. Hence (d) is correct. It may additionally be noted**

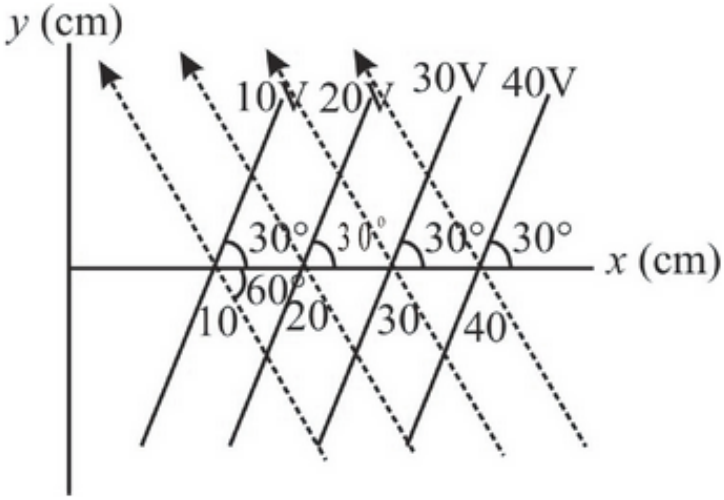
Q.15 Answer:

$\vec{E} = 100(-\hat{i} + \sqrt{3}\hat{j})NC^{-1}$

Solution:

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Dotted lines will give the direction of the electric field as in figure.



Distance between equipotential lines

$\Delta r = 10 \cos 60^\circ = 5 \text{ cm}$

$|\vec{E}| = \left|\frac{\Delta V}{\Delta r}\right| = \frac{10}{5/100} = 200 \text{Vm}^{-1}$

$\vec{E} = |\vec{E}|\hat{u} = 200\left[-\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j}\right]$

$\vec{E} = 200\left[-\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right]$

$\vec{E} = 100[-i + \sqrt{3}j]NC^{-1}$

Q.16 Answer:

Only (ii) and (iii)

Solution:

+Q charge will flow into earth.

$\text{v}_C = \frac{kQ}{r} - \frac{KQ}{b}$

$\text{v}_A = \frac{kQ}{a} - \frac{KQ}{b}$

Potential of B is zero.

Q.17 Answer:

$-67.5\varepsilon_0 \text{C}$

Solution:

To find the charge enclosed, we need the flux through the parallolepiped:

$\varphi_1 = \text{AE}_1 \cos 60^\circ$

$= (0.0500 \text{ m})(0.0600 \text{ m})\left(2.50 \times 10^4 \text{NC}^{-1} \cos 60^\circ\right.$

$= 37.5 \text{Nm}^2\text{C}^{-1}$

$\varphi_2 = \text{AE}_2 \cos 120^\circ$

$= (0.0500 \text{ m})(0.0600 \text{ m})\left(7.00 \times 10^4 \text{NC}^{-1} \cos 60^\circ\right.$

$= -105 \text{Nm}^2\text{C}^{-1}$

So, the total flux is

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$\varphi = \varphi_1 + \varphi_2 = (37.5 - 105)\mathrm{Nm^2C^{-1}} = -67.5\mathrm{Nm^2C^{-1}}$

$q = \varphi\varepsilon_0 = \left(-67.5\mathrm{Nm^2/C}\varepsilon_0 = -5.97 \times 10^{-10}\mathrm{C}\right.$

There must be a net charge (negative) in the parallelopiped since there is a net flux flowing into the surface. Also, there must be an external field or all lines would point toward the slab.

Q.18 Answer:

q/ε_0

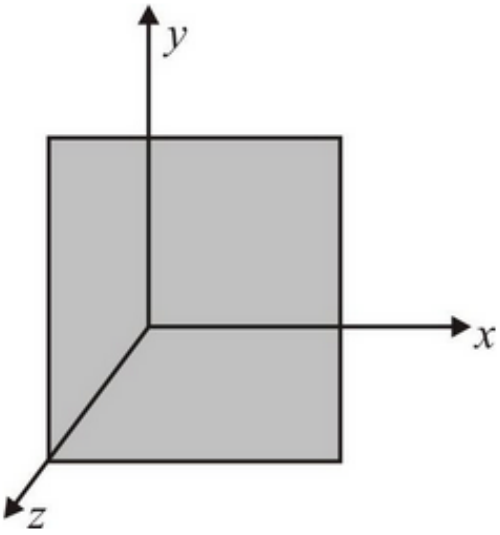
Solution:

Net charge on the conductor will be zero. So, net charge inside S_1 will be the charge on the rod. hence, flux through S_1 is q/ε_0

Q.19 Answer:

$\frac{q}{24\varepsilon_0}$

Solution:



Imaging a charge q at the center of a cube of edge length $2\ \mathrm{L}$ (figure). Then:

$\phi = \frac{q}{\varepsilon_0}$

Here, the square is one 24th of the surface area of the imaginary cube, so it intercepts $1/24$ of the flux. That is,

$\phi = \frac{q}{24\varepsilon_0}$

Q.20 Answer:

Electric flux entering into sphere and leaving the sphere are same

Solution:

In an electric dipole, the flux coming out from positive charge is equal to the flux coming in at negative charge i.e. total charge on sphere $= 0$. From Gauss law, total flux passing through the sphere $= 0$

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Q.21 Answer:

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Solution:

$W_{AB} = W_{AC} + W_{CB}$

W_{CB} should be zero, because in moving from C to B, we always move perpendicular to the field. Hence, force applied by field and displacement will be at 90°.

$W_{AC} = -e\,(V_C - V_A)$

$V_C - V_A = -E \times AC$

$= -10 \times 4 = -40$

$W_{AC} = -e \times (-40) = 40e$

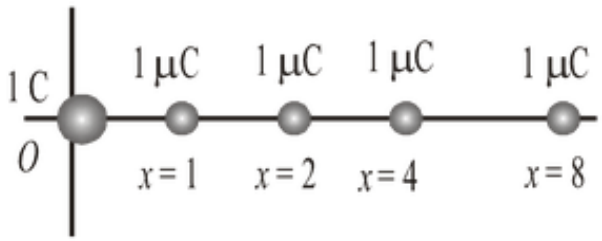
So $W_{AB} = 40eJ = 40\text{eV}$

Q.22 Answer:

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Solution:

The schematic diagram of distribution of charges on x-axis is shown in figure below:



Total force acting on 1C charge is given by

$$F = \frac{1}{4\pi\varepsilon_0}\left[\frac{1\times1\times10^{-6}}{(1)^2} + \frac{1\times1\times10^{-6}}{(2)^2}\ \ \frac{1\times1\times10^{-6}}{(4)^2} + \frac{1\times1\times10^{-6}}{(8)^2} + \ldots.\infty\right]$$

$= \frac{10^{-6}}{4\pi\varepsilon_0}\left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \ldots.\infty\right)$

$= 9\times10^9\times10^{-6}\left(\frac{1}{1-\frac{1}{4}}\right)$

$= 9\times10^9\times10^{-6}\times\frac{4}{3} = 9\times\frac{4}{3}\times10^3 = 12000\text{ N} = 12\text{kN}$

Q.23 Answer:

—

Solution:

$\int_{-\infty}^{0} -\vec{E}\cdot\vec{dl}$ **= potential at centre of non-conducting ring**

$= \frac{1}{4\pi\varepsilon_0}\times\frac{q}{r} = \frac{9\times10^9\times1.11\times10^{-10}}{0.5} = 2$

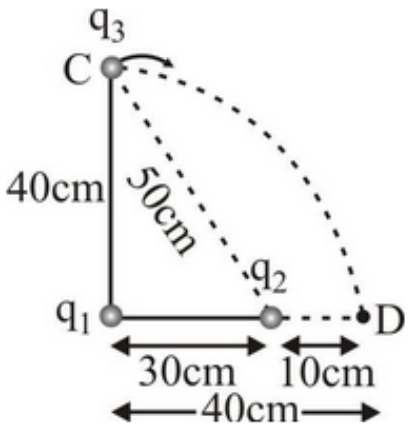
Q.24 Answer:

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Solution:

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Change in potential energy $(\Delta U) = U_f - U_i$



$\Rightarrow \Delta U = \frac{1}{4\pi\varepsilon_0}\left[\left(\frac{q_1 q_3}{0.4} + \frac{q_2 q_3}{0.1}\right) - \left(\frac{q_1 q_3}{0.4} + \frac{q_2 q_3}{0.5}\right)\right]$
$\Rightarrow \Delta U = \frac{1}{4\pi\varepsilon_0}\left[8 q_2 q_3\right] = \frac{q_3}{4\pi\varepsilon_0}\left(8 q_2\right)$
$\therefore k = 8 q_2 = 8$

Q.25 Answer:

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Solution:

Charge enclosed by a cylindrical surface (length $100\ \text{cm}$ **) is** $\mathrm{Q_{enc}} = 100\mathrm{Q}$**. By applying Gauss's law** $\phi = \frac{1}{\varepsilon_0}\left(Q_{enc.}\right) = \frac{1}{\varepsilon_0}(100Q)$

Q.26 Answer:

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Solution:

$$v_{\mathrm{f}} - v_i = -\int_i^f \left(E_x\hat{i} + E_y\hat{j} + E_z\hat{k}\right)\cdot\left(dx\hat{i} + dy\hat{j} + dz\hat{k}\right)$$
$$V_B - V_A = -\left[\int_i^f E_x dx + \int_i^f E_y dy + \int_i^f E_z dz\right]$$
$$V_B - V_A = -\left[\int_1^2 2dx + \int_2^1 3dy\right]$$
$$V_B - V_A = -[2(2-1) + 3(1-2)]$$
$$V_B - V_A = -[2-3] = 1V$$

Hence, $V_A - V_B = -1V$**. Hence, ans is 1**

Q.27 Answer:

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Solution:

Since, the spheres are far apart, charges on them are uniformly distributed, so they may be supposed to be concentrated at the centre for calculating field and potential on and outside the surface.

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$\therefore \quad V_1 = 9 \times 10^9 \frac{3 \times 10^{-8}}{0.06} = 4500 \text{ V}$

$V_2 = 9 \times 10^9 \frac{3 \times 10^{-8}}{0.12} = 2250 \text{ V}$

Let V be the common potential after connection and q_1 and q_2 be the changes on the spheres.

Then $V = 9 \times 10^9 \frac{q_1}{0.06} = 9 \times 10^9 \frac{q_3}{0.12}$

Since total charge remains the same

$2 \times 3 \times 10^{-8} = q_1 + q_2 = \frac{V(0.06+0.12)}{9 \times 10^9}$

or $V = 3000 \text{ V}$

$\therefore \quad q_1 = \frac{3000 \times 0.06}{9 \times 10^9} = 2 \times 10^{-8} C$

$q_2 = \frac{3000 \times 0.12}{9 \times 10^9} = 4 \times 10^{-8} C$

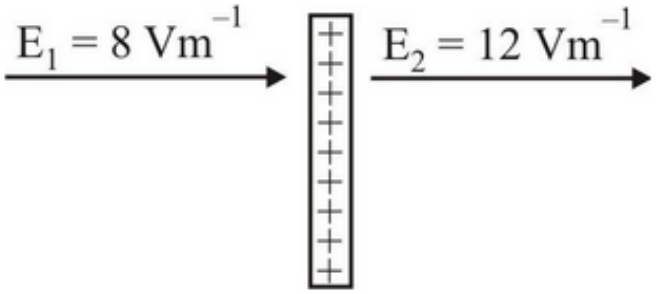
Charge transferred $= 3 \times 10^{-8} - 2 \times 10^{-8}$

$\text{K} = 0.1\text{nC} = 10^{-8}$ **from the small to the large sphere.**

Q.28 Answer:

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Solution:



From the figure it is clear that the plate is placed in an external electric field. Let the electric field due to the plate is E **and** E_0 **be the external electric field.**

$\therefore \text{E}_0 + \text{E} = 12\text{Vm}^{-1}$

$\text{E}_0 - \text{E} = 8\text{Vm}^{-1}$

Solving equation, $\text{E} = 2\text{Vm}^{-1}$ **and**

$\text{E}_0 = 10\text{Vm}^{-1}$

Now, electric field due to plate $= \frac{\sigma}{2\varepsilon_0} = 2$

$\therefore \sigma = 4\varepsilon_0$

Q.29 Answer:

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Solution:

According to Gauss law $\int E \cdot ds = \frac{ql}{\varepsilon_0}$

$\int E \cdot ds = 2\pi rl$; **(E is constant)**

$E \cdot 2\pi rl = \frac{ql}{\varepsilon_0} \Rightarrow E = \frac{q}{2\pi\varepsilon_0 r}$

i.e. $E \propto \frac{1}{r}$

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Q.30 Answer:

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Solution:

$AB + AC = 12 \text{ cm}$ **...(i)**

$AB \cdot AC = 32 \text{ cm}^2$

$\therefore \quad AB - AC = \sqrt{(AB + AC)^2 - 4AB \cdot AC}$

$AB - AC = 4 \quad \ldots (\text{ii})$

From equation (i) and (ii)

$AB = 8 \text{ cm}; AC = 4 \text{ cm}$

Potential energy at point A

$V_A = \frac{1}{4\pi\varepsilon_0} q_1 q_2 \left[\frac{1}{AB} + \frac{1}{AC}\right]$

$= \frac{9\times10^9\times2\times2\times10^{-12}}{10^{-2}} \left(\frac{1}{8} + \frac{1}{4}\right) = 1.35J$

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